

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, m, i, j, k;
```

```
    printf("1WA24CS216\n");
```

```
    printf("Enter number of processes: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter number of resources: ");
```

```
    scanf("%d", &m);
```

```
    int allocation[n][m];
```

```
    int max[n][m];
```

```
    int need[n][m];
```

```
    int available[m];
```

```
    int work[m];
```

```
    int finish[n];
```

```
    int safeSequence[n];
```

```
    printf("\nEnter Allocation Matrix:\n");
```

```
    for(i = 0; i < n; i++) {
```

```
        for(j = 0; j < m; j++) {
```

```
            scanf("%d", &allocation[i][j]);
```

```
        }
```

```
    }
```

```
    printf("\nEnter Maximum Demand Matrix:\n");
```

```
    for(i = 0; i < n; i++) {
```

```
        for(j = 0; j < m; j++) {
```

```
            scanf("%d", &max[i][j]);
```

```
    }  
}
```

```
printf("\nEnter Available Resources:\n");  
for(i = 0; i < m; i++) {  
    scanf("%d", &available[i]);  
    work[i] = available[i];  
}
```

```
for(i = 0; i < n; i++) {  
    for(j = 0; j < m; j++) {  
        need[i][j] = max[i][j] - allocation[i][j];  
    }  
}
```

```
for(i = 0; i < n; i++) {  
    finish[i] = 0;  
}
```

```
int count = 0;
```

```
while(count < n) {  
    int found = 0;
```

```
    for(i = 0; i < n; i++) {
```

```
        if(finish[i] == 0) {
```

```
            int possible = 1;
```

```
            for(j = 0; j < m; j++) {
```

```

        if(need[i][j] > work[j]) {
            possible = 0;
            break;
        }
    }

    if(possible) {

        for(k = 0; k < m; k++) {
            work[k] += allocation[i][k];
        }

        safeSequence[count] = i;
        count++;
        finish[i] = 1;
        found = 1;
    }
}

if(found == 0)
    break;
}

int safe = 1;

for(i = 0; i < n; i++) {
    if(finish[i] == 0) {
        safe = 0;
        break;
    }
}

```

```
}

if(safe) {
    printf("\nSystem is in a safe state.\n");
    printf("Safe sequence is: ");

    for(i = 0; i < n; i++) {
        printf("P%d", safeSequence[i]);

        if(i != n - 1)
            printf(" -> ");
    }

    printf("\n");
}

else {
    printf("\nSystem is not in a safe state.\n");
}

return 0;
}
```

```
C:\Users\Admin\Desktop\1W x + v
1WA24CS216
Enter number of processes: 5
Enter number of resources: 3

Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2

Enter Maximum Demand Matrix:
7 5 3
3 2 2
9 0 2
2 2 2
4 3 3

Enter Available Resources:
3 3 2

System is in a safe state.
Safe sequence is: P1 -> P3 -> P4 -> P0 -> P2

Process returned 0 (0x0) execution time : 51.124 s
Press any key to continue.
```